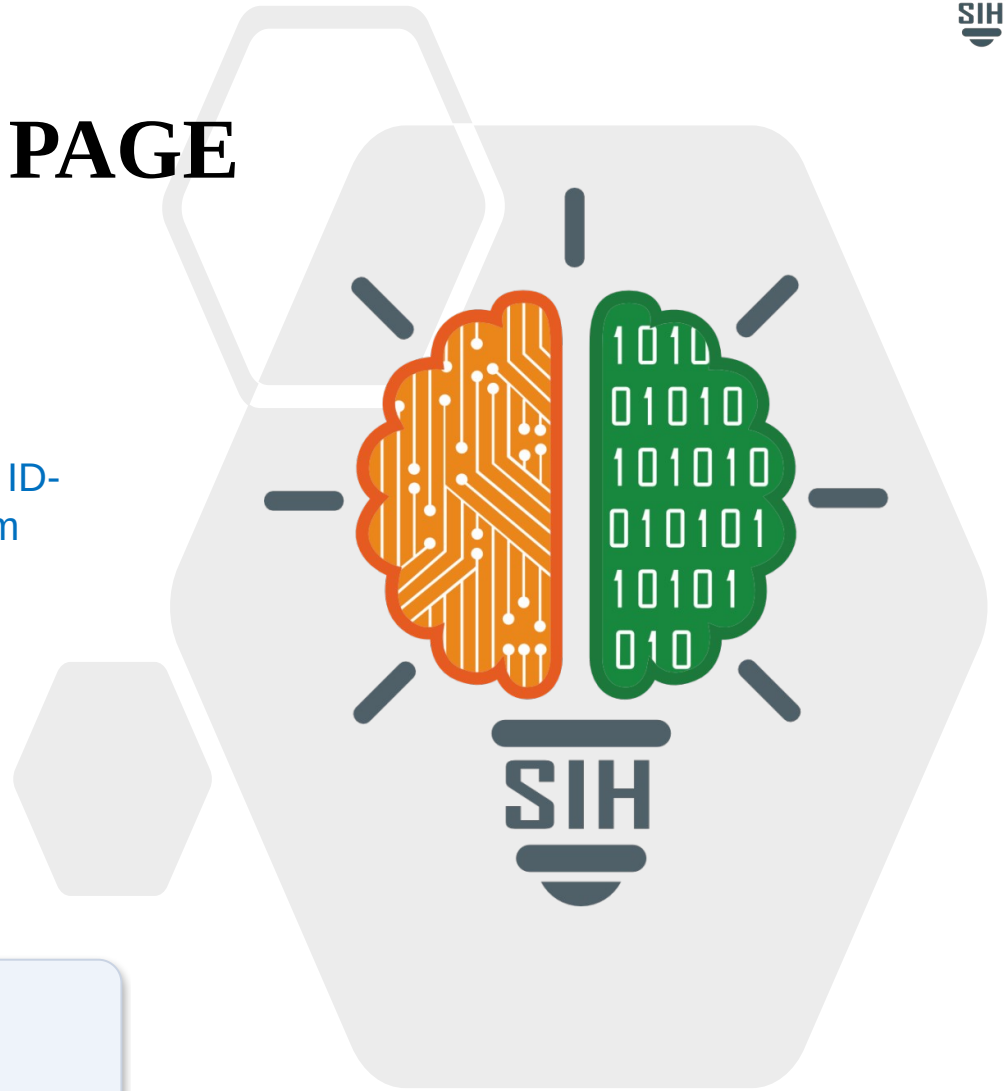
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TITLE PAGE



- **Problem Statement ID –** SIH26263
- **Problem Statement Title-** Development of an APAAR ID-Based Student Data Management and Verification Platform for AICTE-Approved Institutions
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal)** AI Riders

VIDYA PRAMAAN

One student. One APAAR ID. One verifiable record.

Working prototype: https://vidya-pramaan-sih26263.onrender.com

AI Riders

VIDYA PRAMAAN

APAAR-linked student data, validation & verification platform for AICTE

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Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution



How it addresses the problem

✖ Only totals reach AICTE; AISHE is 2–3 years late	✔ Student-level records keyed on APAAR, live
✖ Excess admission = 5× fee fine per student, found late	✔ Seat #61 of 60 blocked at upload and at submit
✖ Same student at 2 colleges; agent-filled admissions	✔ Cross-college match + shared-contact rings
✖ Records can't be verified at source	✔ Signed QR (checks offline) + consent for more

Innovation and uniqueness of the solution

Twin-safe matching
Mohd = Mohammed, but Rohit ≠ Mohit

Ghost-admission map
1 phone, 8 students, 2 colleges: flagged

Decisions kept
reason required; never re-flagged

Signed checkpoints
even a full history rewrite is caught

Rules as data
2026-27 caps by level, each with its source

APAAR on degrees
university print feed with verification QR

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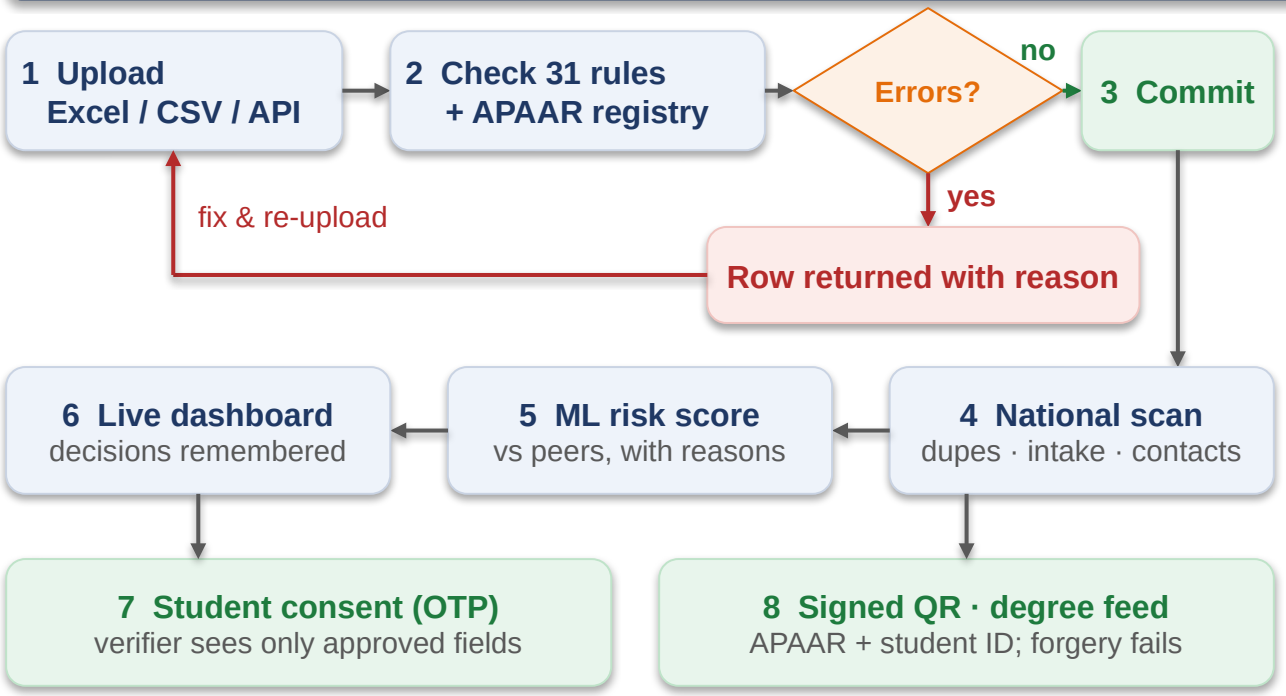
TECHNICAL APPROACH

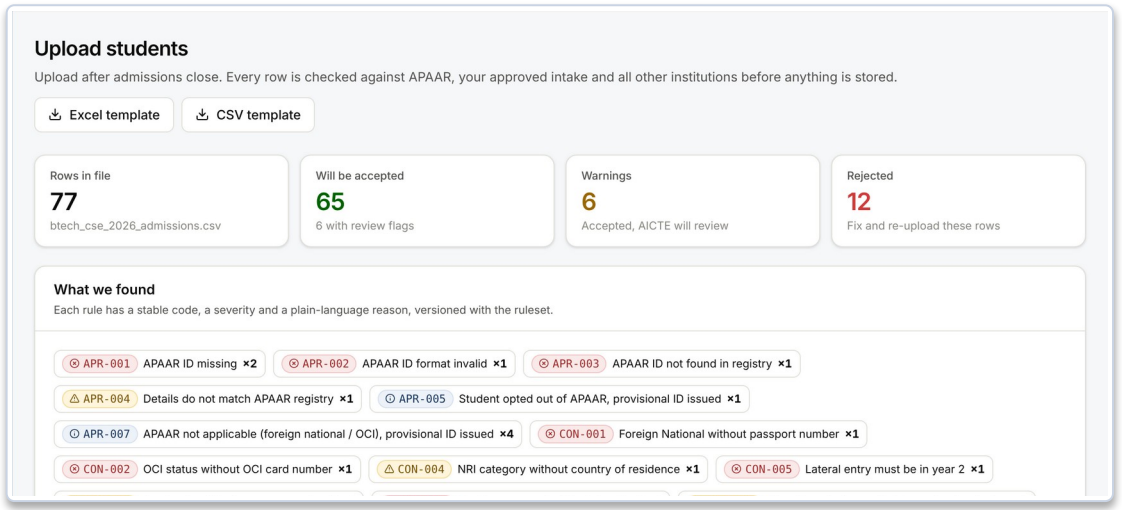
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❖ Technologies to be used (e.g. programming languages, frameworks, hardware)

- **Python · FastAPI** — REST · OpenAPI
- **React · TypeScript** — installable PWA
- **PostgreSQL 16** — SQLite in the demo
- **scikit-learn** — Isolation Forest
- **Name matcher** — twin-safe · Indic
- **Ed25519 · W3C** — signed QR + audit

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)





Working prototype: 77-row upload → 65 accepted, 12 returned with rule codes

Try it live: https://vidya-pramaan-sih26263.onrender.com
7 demo roles · synthetic data · installable on phones

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FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea

0.79 / 0.92
held-out precision / recall (old matcher 0.65 / 0.68)

0 / 80
same-day twins wrongly merged

17/17
planted error types; 0 false flags on 56 clean rows

164 · 57 · 4
planted dual enrolments, identity dupes, contact rings

42/46
top-ranked colleges truly anomalous

58 tests
SQLite + PostgreSQL 16; 7-role browser walkthrough

Planted cases are regression checks on synthetic data, not real-world accuracy · precision / recall come from independent hand-labelled name pairs

Potential challenges and risks	Strategies for overcoming these challenges
No public APAAR API	✔ Registry adapter: mock now → DigiLocker / API Setu after partner onboarding
Students without APAAR (opt-out, FN / OCI)	✔ Provisional ID, never rejected; mother's or guardian's name accepted
Student privacy (DPDP Act; APAAR is voluntary)	✔ Consent per field, masked IDs, no Aadhaar stored, access log
Wrong flags: twins, spelling variants	✔ Twin-safe matcher; officer decides with a reason; decision remembered
Rules change every year by circular	✔ Rules file per academic year, each value with its source
ERP formats, low digital readiness	✔ ERP headers auto-mapped, Excel template, API keys, Hindi, phone app

Roadmap M1–3 pilot: 1 state, 50 colleges → M4–6 API Setu, SSO, STQC audit → M7–12 national **Cost** open source, no licence fees; est. ₹20–25 lakh/yr MeitY-empanelled cloud

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IMPACT AND BENEFITS

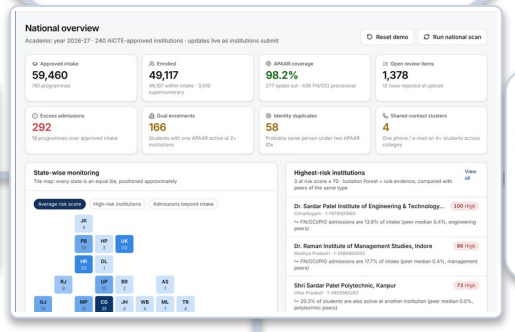
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Potential impact on the target audience

AICTE & regulators
excess & dual enrolment caught at submission

Institutions
one upload; errors and corrections explained

Universities
APAAR + student ID printed on degrees



One national view

Employers
instant QR check, even offline; consent for more

Students
own record; control over who sees it

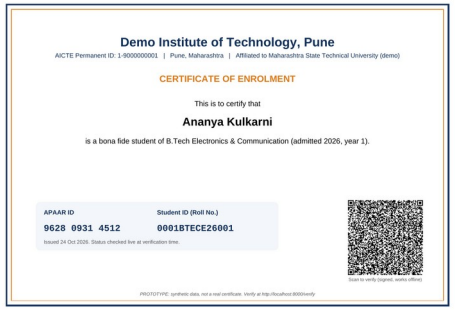
Benefits of the solution (social, economic, environmental, etc.)

Social
fair admissions; no student excluded for opting out

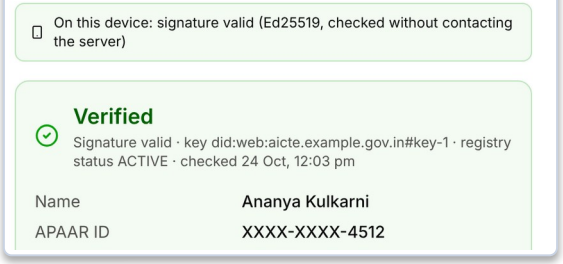
Economic
no licence cost; one upload replaces repeated returns

Environmental
paperless returns and verification; no courier checks

Governance
audited evidence pack for review and show-cause



Certificate: APAAR ID + student ID + QR



QR scan: verified, also on the device

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RESEARCH AND REFERENCES

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❖ Details / Links of the reference and research work

Why not just use the systems that exist?

	NAD / DigiLocker	ABC	AISHE	Vidya Pramaan
Student-level admissions vs approved intake	—	—	aggregate, 2–3 yr lag	✔ live
Same student at two colleges; ghost admissions	—	—	—	✔
Supernumerary caps by programme and year	—	—	—	✔
Consent-based verification for employers	✔	—	—	✔ field-level
APAAR + student ID feed for degree printing	issuer uploads	credits only	—	✔

Vidya Pramaan consumes APAAR and feeds NAD / ABC and AISHE: no new ID, no parallel credential store, no invented APAAR checksum (none is published), and no student rejected for opting out.

References

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5. AISHE 2022-23 & 2023-24 reports (released 2026) aishe.gov.in
6. APAAR consent and opt-out: MoE guidance; Supreme Court proceedings 2026 apaar.education.gov.in
7. Digital Personal Data Protection Act 2023; DPDP Rules 2025 meity.gov.in
8. W3C Verifiable Credentials; RFC 8032 (Ed25519) w3.org · rfc-editor.org
9. Liu et al. (2008) Isolation Forest; IDinsight (2021) matching Indian names doi.org/10.1109/ICDM.2008.17 · idinsight.org

Live prototype
https://vidya-pramaan-sih26263.onrender.com
Synthetic data only · 7 demo roles · works on phones

Source code & tests
https://github.com/Pulkitkh/VidyaPramaan
58 automated tests + browser walkthrough · MIT licence